ModelIt!

Student Activity Pack

Whose Air Is This?

Why Kids Near Freeways Have Higher Asthma Rates

Grade 5 | NGSS 5-ESS3-1 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Do all neighborhoods have the same air quality? Explain.

2 What causes air pollution in a neighborhood?

3 Children near freeways have higher asthma rates. What explains this?

4 Is it fair that some neighborhoods have worse air than others?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Vehicle Traffic Density | Wind Speed | Air Pollution Concentration | Respiratory Health Impact

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Normal Day

Moderate traffic and wind

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Rush Hour + Still Air

Traffic 90%, Wind 10%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Holiday/Weekend

Traffic 20%, Wind moderate

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Green Infrastructure / Trees — Trees capture particulate matter

Electric Vehicle Percentage — EVs produce no tailpipe emissions

Temperature Inversion — Warm air trapping pollution near ground

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How do vehicle traffic and wind patterns determine what a community breathes? Use evidence from your simulation to support your answers.

1. **Why does high wind reduce respiratory health impact in your model?**

2. **Why might a community near a freeway NOT be able to solve this alone?**

3. **What is one solution communities are using to improve air quality?**

4. **Is this a science problem, a justice problem, or both?**

5. **What role should science play in community advocacy?**

8 STEM Challenge: Design an Air Quality Solution

ModelIt!

YOUR ENGINEERING MISSION

Propose a science-based solution to improve air quality near a freeway.

THE SCENARIO

A community near a freeway has high asthma rates. They can't move the freeway. Design a solution to help.

Engineering Design Thinking

1. WHAT SPECIFIC INTERVENTION WILL YOU PROPOSE?

2. HOW DOES IT CONNECT TO YOUR MODEL?

3. WHAT ARE THE COSTS AND BENEFITS?

4. WHO NEEDS TO TAKE ACTION FOR IT TO HAPPEN?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Environmental Health Scientists and Air Quality Monitors protect communities from pollution. They earn $60,000-$95,000/year.